

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer on a provider-facing **SMART on FHIR** clinical decision support app (Next.js + Firebase/GCP) that surfaces covered, guideline-backed medication alternatives in the EHR prescribing flow to prevent prior authorization.
- Extending a **Firebase Cloud Functions** backend that builds alternatives from **RxNorm** normalization, formulary/prior-auth logic, and PubMed/guideline evidence, with **Gemini** constrained to grounded synthesis while the backend deterministically controls eligibility and coverage.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's CCR (Contextualized Construct Representations) text-analysis platform (Python, **FastAPI**, sentence-transformers, React) with full technical ownership delegated by the PI, productionizing a published NLP method into a self-serve web tool for researchers.
- Built a YAML model registry, a versioned construct library (**99 constructs from 38 questionnaires**), and a structured data-quality warning system; grew the backend suite to **40 tests** (hermetic CI) and set up **Claude Code** multi-agent repo governance (custom review agents, protective hooks).

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack internal ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Served as primary technical point of contact for the eTriage backend; ran structured knowledge transfer that onboarded a second engineer to on-call readiness in **~6 weeks**, and was selected for a COO-sponsored **GitHub Copilot** productivity pilot across tech-debt PRs, migrations, and security-finding triage.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service turning open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and **cell-level provenance**; hardened with **150+ tests**.

align-ops – LLM Post-Training + Serving Ops PyTorch, TRL (SFT / DPO / GRPO), vLLM, FastAPI, Docker

- Post-trained Qwen2.5-1.5B through SFT (QLoRA), DPO, and GRPO, lifting held-out task accuracy **54% → 72%** with a **71% preference win-rate** over base across 500 prompts; served via vLLM behind a FastAPI gateway, where async micro-batching and Redis caching delivered **1.9x throughput**.

KG2RAG-Enhanced – Multi-Hop QA on HotpotQA (GitHub) Python, Ollama / LLaMA-3, sentence-transformers, RRF

- Built the multi-view retrieval component of a KG-guided RAG pipeline (query-side decomposition, RRF fusion, cross-encoder reranking, MMR), lifting supporting-fact recall **+2.68%** over the KG²RAG baseline on a 4,905-question evaluation.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, TypeScript, JavaScript, C++, SQL, Bash, MongoDB, PostgreSQL, SQLite, Redis

LLM & Agents: LLM agents, RAG, MCP, prompt & tool design, OpenAI / Groq / Gemini APIs, LLM evaluation, provenance

ML & Fine-Tuning: PyTorch, Hugging Face TRL (SFT / DPO / GRPO), QLoRA, vLLM, Transformers, sentence-transformers

Frameworks / Tools: FastAPI, Node.js, Express, React, Docker, Git, GitHub Actions, Prometheus, Grafana, Claude Code

Cloud / Platforms: AWS (KMS, S3), GCP, Azure, Linux, MongoDB Atlas

Open Source & Community

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.